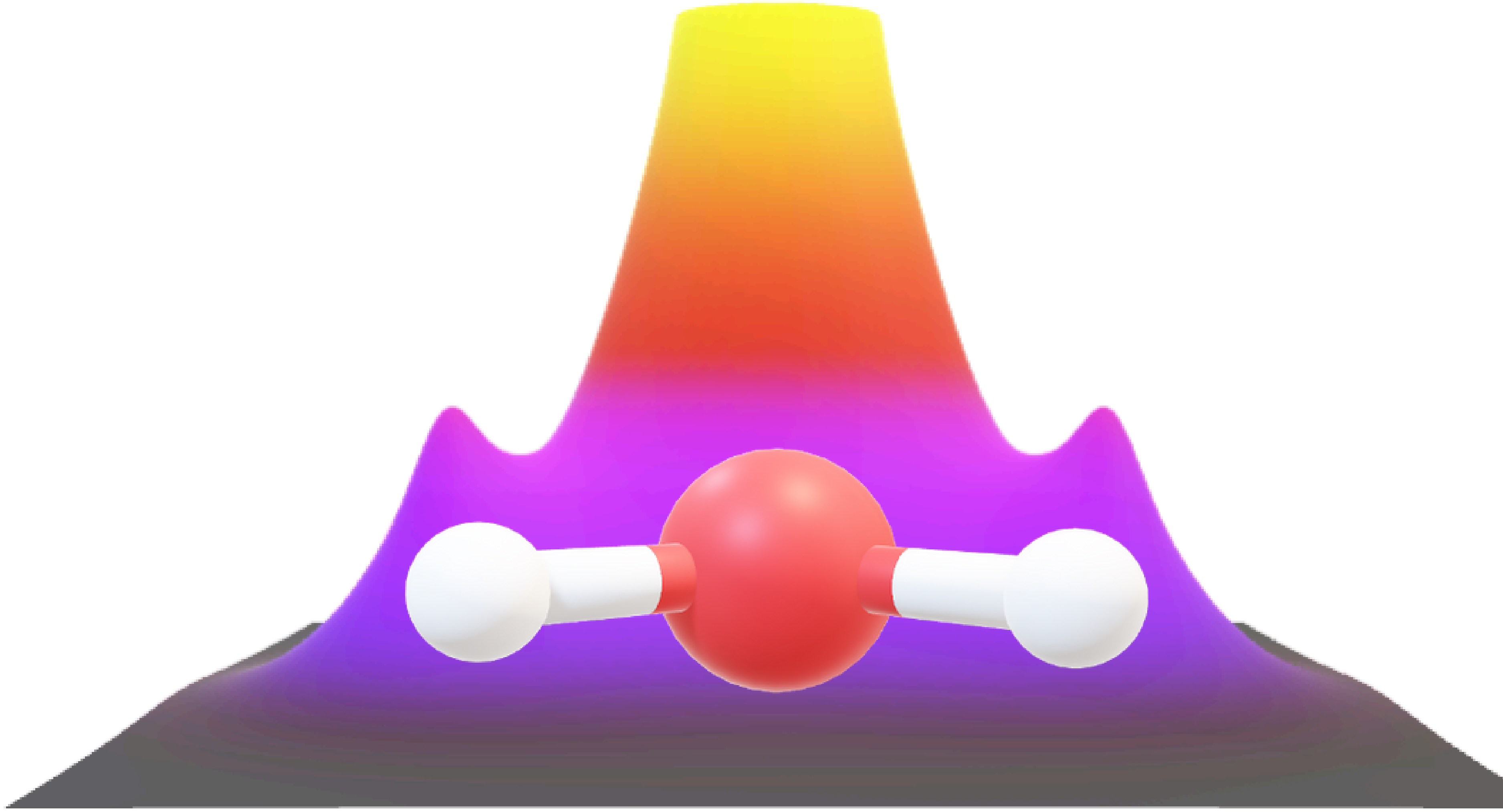
Charge density topological analysis: TOPOND



MSSC25

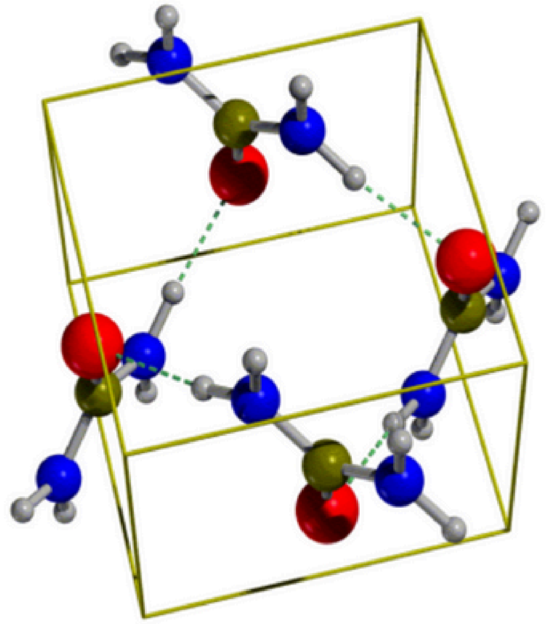
Overview

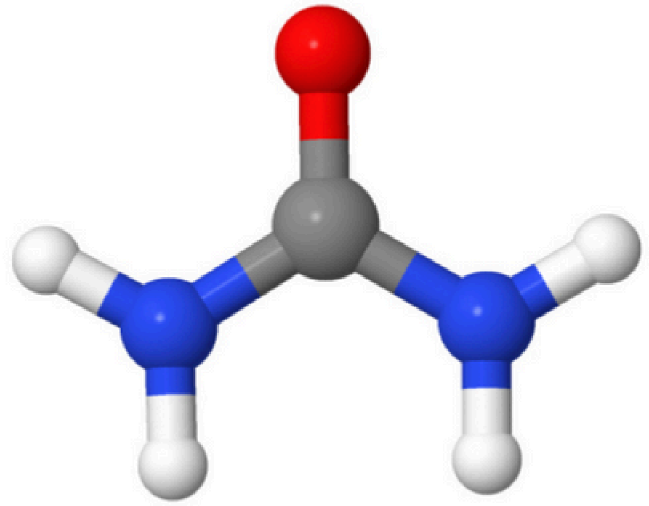
- **Topological analysis of the electron** (charge) density, $\rho(r)$, according to the Quantum Theory of Atoms in Molecules and Crystals (**QTAIMC**) by Richard Bader
- The module in the CRYSTAL code to do this is named **TOPOND**
- Accessible using the keyword **TOPO** (using *properties*)

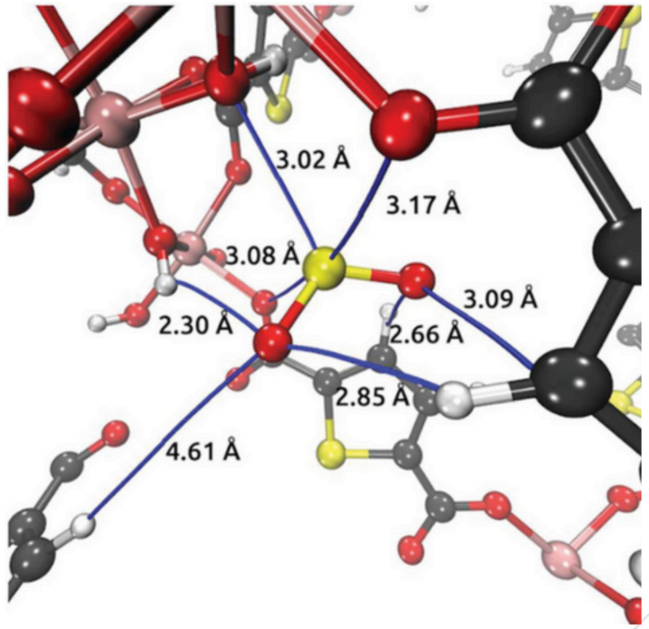
- ***Section I***: Topological analysis of $\boldsymbol{\rho}$ (keyword **TRHO**). It provides the Critical Points, ρ_{CP}: the points where the gradient of the density, $\nabla\rho$, vanishes.
- ***Section II***: Topological analysis of the Laplacian ($\boldsymbol{\nabla^2\rho}$) of the electron density (keyword **TLAP**).
- ***Section III***: Determination of **atomic basins** and their associated properties. (keyword **ATBP**)
- ***Section IV***: 2-dimensional plots data (keywords **PL2D**)

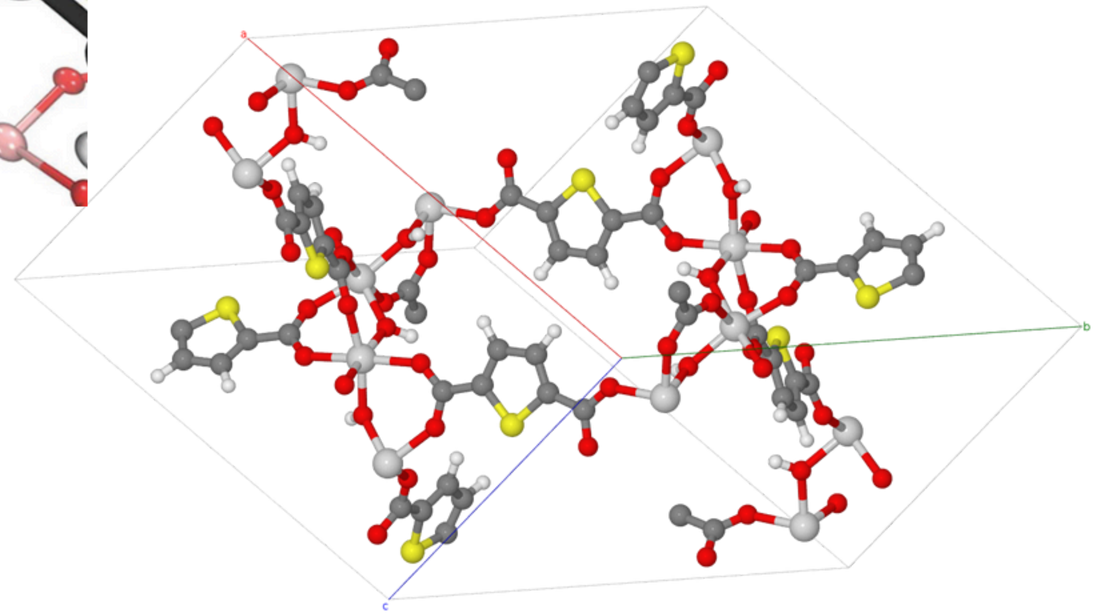
Scope of the tutorial

- To perform a complete topological characterization of the **molecular crystal of Urea**, in which covalent as well as hydrogen and van der Waals interactions are present.
- To compare this result with the corresponding properties of the **Urea molecule**, in order to emphasize the modification occurring on the electron density (and related properties) when the crystalline lattice is formed.
- **Extra**: Analysis of a **molecule adsorption inside a MOF** porous.



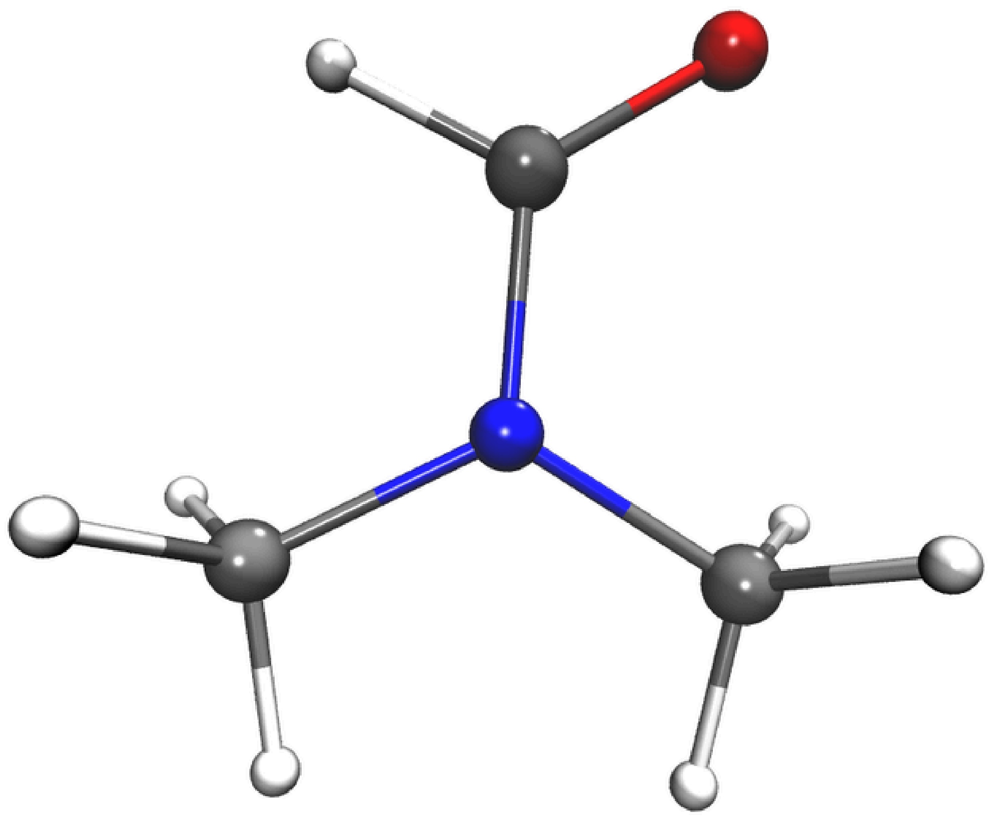




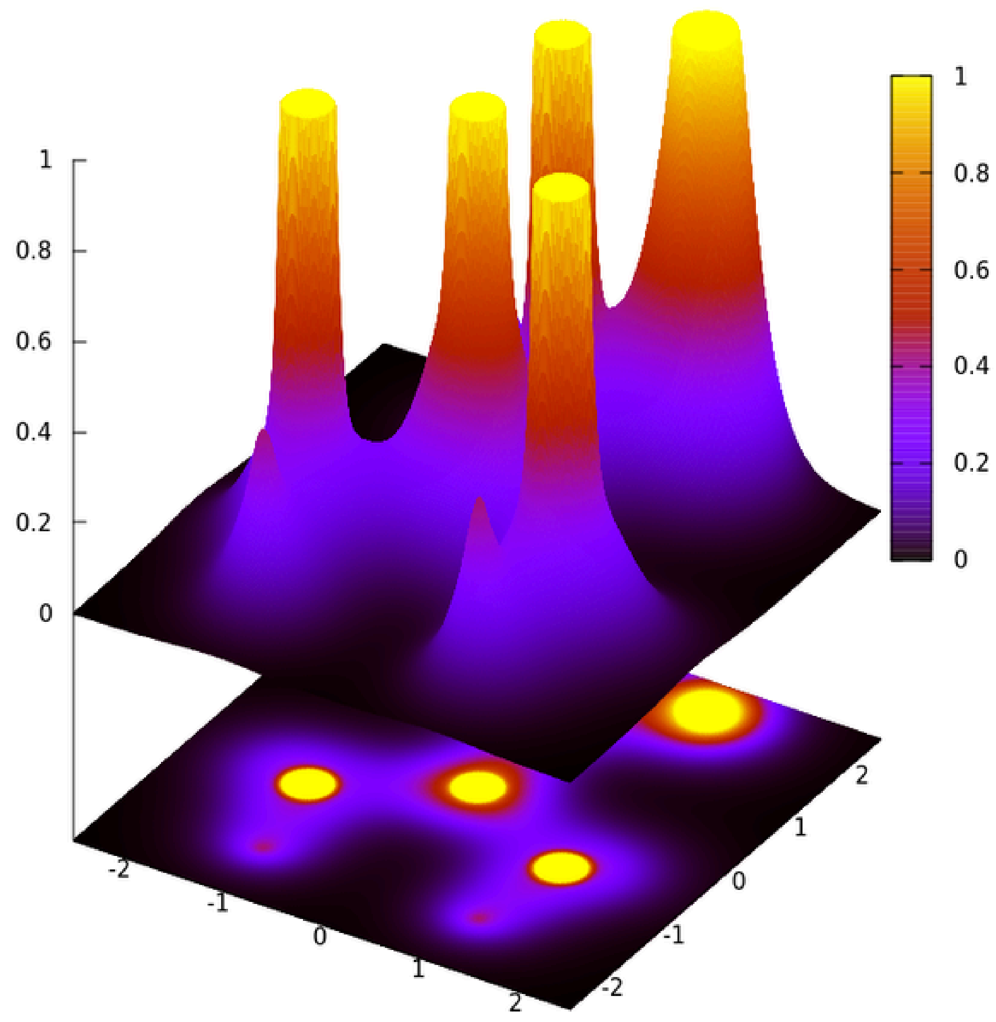


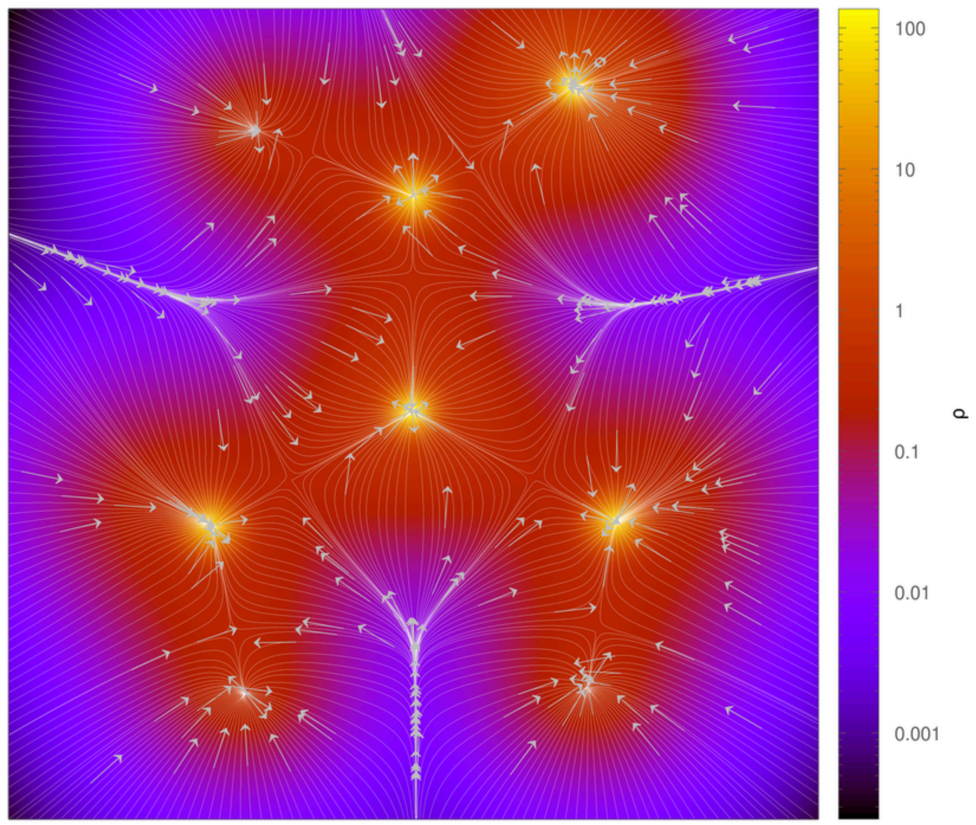
Background

- TOPOND implements Bader's QTAIMC.
- Given the *topology/shape/distribution* of ρ (experimental observable), can we define what is an atom/bond/interaction?



DMF



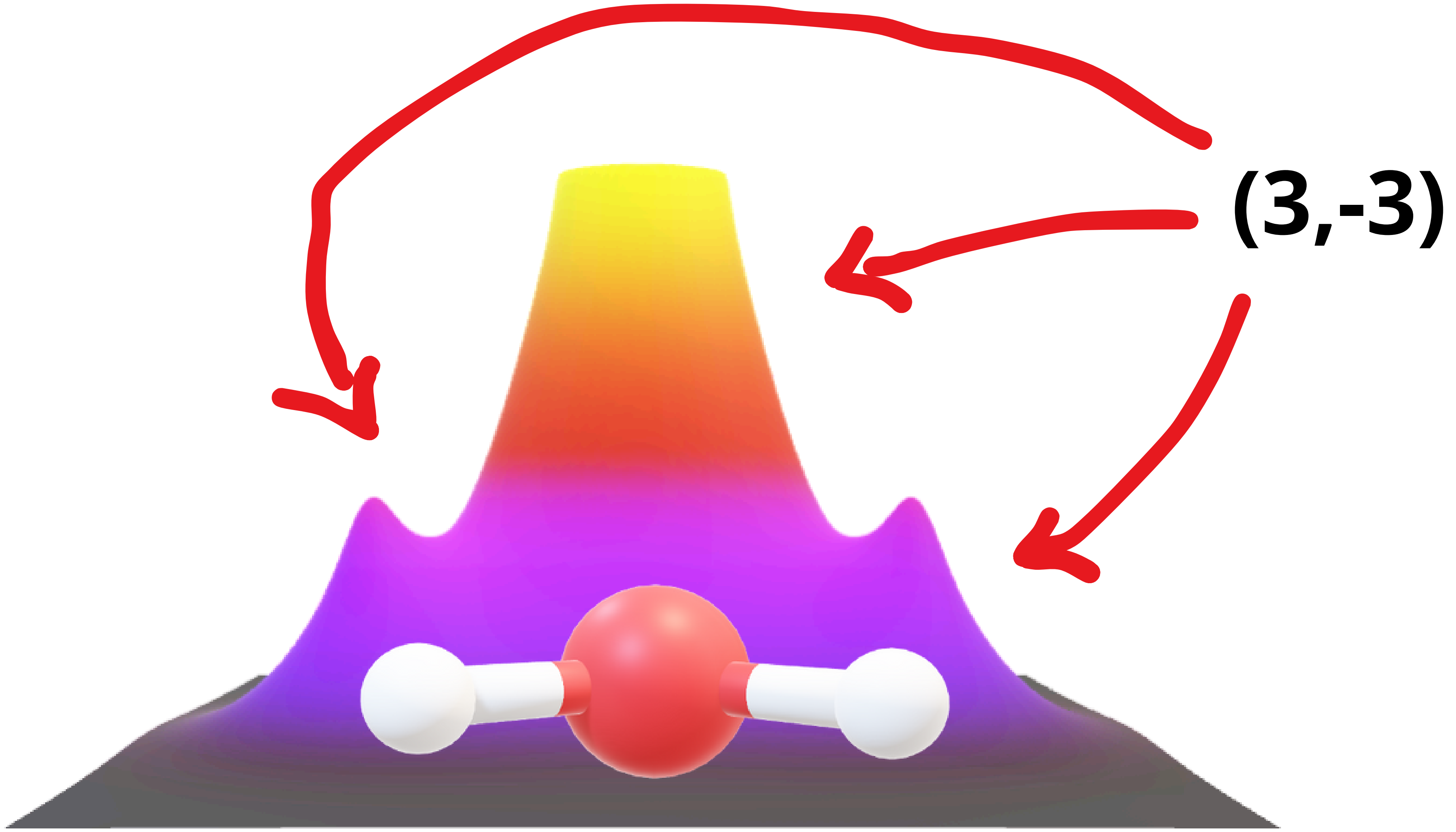


Topological Analysis of the Electron Density

- Keyword **TRHO**
- It provides the **Critical Points**, ρ_{CP}: the points where the gradient of the density, $\nabla\rho$, vanishes.
- ρ_{CP} may be **classified** in terms of their type **(r,s)** where **r** is the *rank* (# of non-zero eigenvalues of the Hessian matrix) and **s** the *signature* (sum of signs of the eigenvalues).
- Stable structures are characterized by all their critical points of ρ being of rank 3. Signature also indicates number of positive/negative curvatures at the point.
- The CPs may be associated with chemically recognizable entities in a crystal, as in the **table**.

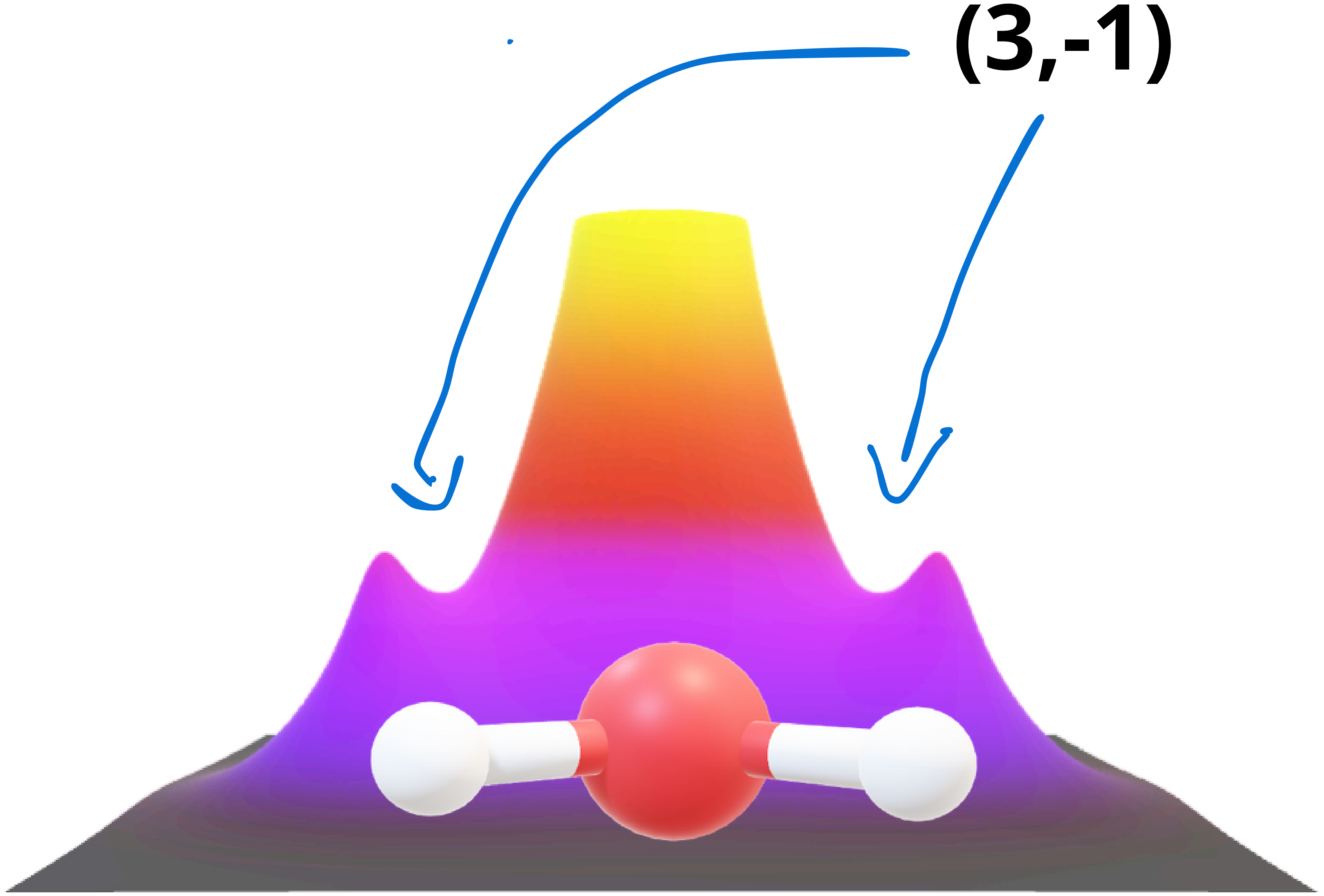
(r,s)	**CP Type**	**Entity**
(3,-3)	Maximum	Nucleus/ NNA
(3,-1)	Saddle	Bond
(3,+1)	Saddle	Ring
(3,+3)	Minimum	Cage

Topological Analysis of the Electron Density



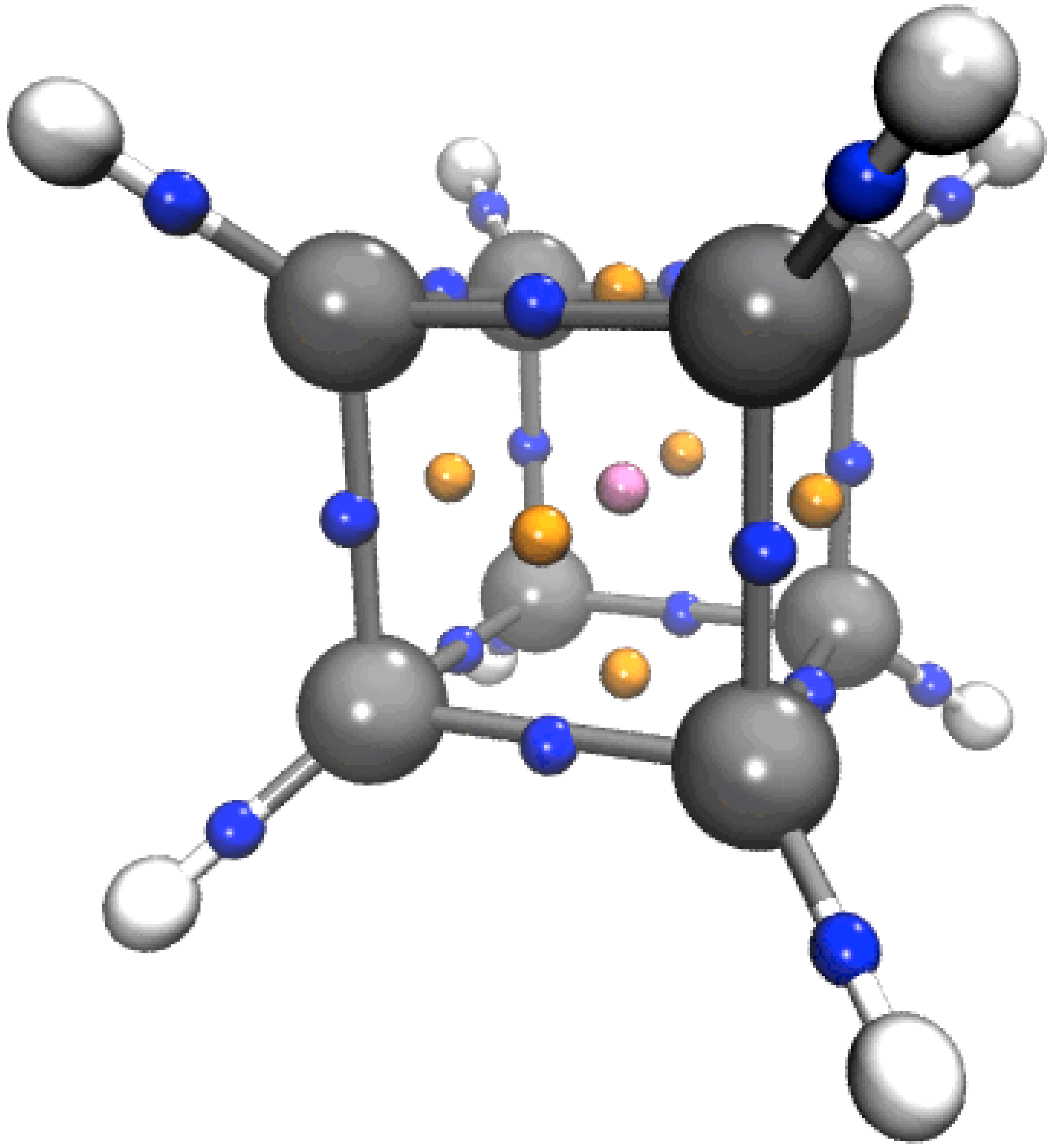
(r,s)	CP Type	Entity
(3,-3)	Maximum	**Nucleus/ NNA**
(3,-2)	Saddle	**Bond**
(3,+1)	Saddle	**Ring**
(3,+3)	Minimum	**Cage**

Topological Analysis of the Electron Density



(r,s)	CP Type	Entity
(3,-3)	Maximum	**Nucleus/ NNA**
(3,-2)	Saddle	**Bond**
(3,+1)	Saddle	**Ring**
(3,+3)	Minimum	**Cage**

Topological Analysis of the Electron Density



(r,s)	CP Type	Entity
(3,-3)	Maximum	**Nucleus/ NNA**
(3,-2)	Saddle	**Bond**
(3,+1)	Saddle	**Ring**
(3,+3)	Minimum	**Cage**

TRHO input

TOPO	Keyword to call TOPOND
TRHO	Keyword for TRHO functions
-1	IAUTO (Search Strategy): Fully automated chain-like search strategy for all kind of CPs. By chain-like, it is meant that (3,-3), (3,-1), (3,+1) and (3,+3) CPs are separately searched for, according to this listed sequence.
1,0,1,0,20,10,7,0	8 parameters related to IAUTO=-1. Explained in next slide
END	Keyword to close the input

TRHO input

- **IAUTO=-1: fully automated** search strategy for all kinds of critical points.
- **Search region** defined as the union of the regions of space obtained by building-up molecular clusters centered at each of the non-equivalent atom (NEA) of the unit cell.
- Bond paths lengths and their termini are evaluated. Kinetic energy densities, Virial density and Laplacian are evaluated at each **bond critical points**.

IAUTO=-1 options:

1. IEXT (1): properties (kinetic energy densities, virial density, Electron Localization Function) depending on the non-diagonal elements of the first-order density matrix are also evaluated at each unique CP
2. ICRIT (0): all kinds of CPs are searched for
3. IBPAT (1): Bond paths (or AILs) lengths and termini are evaluated numerically for each unique (3,-1) CP.
4. PRINT (0): normal printing (no debug)
5. NSTEP (20): maximum number of EF steps along each search.
6. NNB (10): maximum number of symmetry-related stars of atoms to be included in the clusters generated around each NEA
7. RMAX (7): maximum radius of the clusters.
8. TH (0): the default threshold value (5 Å) is used in the (3,-1) CP search stage

TRHO output

```
********************
SEARCH OF (3,-1) CPS
********************

CP N.      6
*********

CP TYPE                          :  (3,-1)
COORD(AU)  (X  Y  Z)             :  0.0000E+00  3.9475E-18  8.3094E-01
PROPERTIES (RHO,GRHO,LAP)        :  3.8362E-01  1.0193E-16 -6.6445E-01
KINETIC ENERGY DENSITIES (G,K)   :  4.1701E-01  5.8312E-01
VIRIAL DENSITY                   : -1.0001E+00
ELF(PAA)                         :  6.6040E-01

SPECTRAL DECOMP. OF THE HESSIAN OF RHO

EIGENVALUES (L1 L2 L3)           : -1.0085E+00 -8.9598E-01  1.2400E+00
EIGENVECTORS                     :  0.0000E+00  1.0000E+00  0.0000E+00
                                    1.0000E+00  0.0000E+00  0.0000E+00
                                    0.0000E+00  0.0000E+00  1.0000E+00

ELLIPTICITY                      :  1.2561E-01

CLUSTER OF ATOMS AROUND THE CP
NEW  OLD    AT. NU.       COORD.(AU)              DISTANCE (ANG)
   1     1       6     0.000   0.000   0.000           0.440
   2     2       8     0.000   0.000   2.384           0.822
   3     4       7     0.000  -2.170  -1.322           1.618
   4     3       7     0.000   2.170  -1.322           1.618
   5     6       1     0.000  -3.830  -0.383           2.126
   6     5       1     0.000   3.830  -0.383           2.126
   7     8       1     0.000  -2.143  -3.222           2.426
   8     7       1     0.000   2.143  -3.222           2.426
```

Some important quantities to check:

- CP type
- CP Coordinates
- Properties and relevant quantities (ρ, $\nabla\rho$, $\nabla^2\rho$)
- Eigenvalues and eigenvectors
- Ellipticity ($\epsilon=\lambda_1/\lambda_2-1$). ϵ close to zero $\rightarrow$ circular ρ distribution
- Length of the chemical bond:
  - Bond Path Length (BPL)
  - Distance A-B (RAB)

```
BPL (ANG) ; RAB ; (BPL/RAB)    :  1.2614E+00  1.2614E+00  1.0000E+00

CP N.      7
*********
```

TRHO Classification of bonds

- Kinetic energy densities (G,K), Virial density and Laplacian are evaluated at each bond critical points.

- Different classifications based on these quantities are available:

Chemical bonding in crystals: new directions **409**

Table 4. Classification of atomic interactions.

The dichotomous classification[a] based on the sign of $\nabla^2\varrho_b$

Property	Shared shell, $\nabla^2\varrho_b < 0$ Covalent and polar bonds	Closed-shell, $\nabla^2\varrho_b > 0$ Ionic, H-bonds and vdW molecules
λ_1	$\lambda_{1,2}$ dominant; $\lvert\lambda_{1,2}\rvert/\lvert\lambda_3 > 1$	λ_3 dominant; $\lvert\lambda_{1,2}\rvert/\lvert\lambda_3 \ll 1$
VSCC[b]	The VSCCs of the two atoms form one continous region of charge concentration	$\nabla^2\varrho > 0$ over the entire interaction region. The spatial display of $\nabla^2\varrho$ is mostly atomic-like
ϱ_b	Large	Small
Energy lowering	By accumulating ϱ in the interatomic region	Regions of dominant $V(\mathbf{r})$ are separately localized within the boundaries of interacting atoms
Energy components	$2G_b < \lvert V_b\rvert$; $G_b/\varrho_b < 1$; $G_{b\parallel} \ll G_{b\perp}$; $H_b < 0$	$2G_b > \lvert V_b\rvert$; $G_b/\varrho_b > 1$, $G_b \gg G_{b\perp}$; H_b any value

Bond polarity is increasing - - - - - - - - - - - →

← - - - - - - - - - - - *Bond covalency is increasing*

The classification[c] based on the adimensional $\lvert V_b\rvert/G_b < 1$ ratio

Shared shell (SS) $\lvert V_b\rvert/G_b > 2$	Transit region, incipient covalent bond formation $1 < \lvert V_b\rvert/G_b < 2$	Closed-shell (CS) $\lvert V_b\rvert/G_b < 1$
$H_b < 0$; $\nabla^2\varrho_b < 0$	$H_b < 0$; $\nabla^2\varrho_b > 0$	$H_b > 0$; $\nabla^2\varrho_b > 0$
Bond degree (BD) $= H_b/\varrho_b \equiv$ Covalence degree (CD)	BD $\equiv$ CD	BD $\equiv$ Softness degree (SD)
BD large and negative	BD negative and smaller in magnitude than for SS interactions	SD positive and large
The larger is $\lvert$BD$\rvert$ the more covalent is the bond	BD Approaching zero at the boundary with CS region	The larger is SD the weaker and closed-shell in nature is the bond

TRHO Classification of bonds

- Kinetic energy densities (G,K), Virial density and Laplacian are evaluated at each bond critical points.

- Different classifications based on these quantities are available:

The classification[d] based on the atomic valence shell and on both the local (*bcp*) and integral properties

	ϱ_b	$\nabla^2\varrho_b$	G_b/ϱ_b	H_b/ϱ_b	δ(A,B)	$\oint_{AB} \varrho(\mathbf{r}_s)\, d\mathbf{r}_s$
Bonds between light atoms						
Open-shell (covalent bonds); e.g. C–C, C–H, B–B	Large	$\ll 0$	<1	$\ll 0$	$\sim$ Formal bond order	Large
Intermediate interactions (polar bonds, donor-acceptor bonds; e.g. C–O, H_3B–CO	Large	any value	≥ 1	$\ll 0$	$<$ Formal bond order	Large
Closed-shell (ionic bonds, HBs, van der Waals interactions; e.g. LiF, H...O, Ne...Ne	Small	>0	≥ 1	>0	~ 0	Small
Bonds between heavy atoms						
Open-shell (e.g. Co–Co)	Small	~ 0	<1	<0	Formal bond order (unless bond delocalisation occurs)	Medium/large
Donor acceptor (e.g. Co–As)	Small	>0	~ 1	<0	$<$ Formal bond order	Medium/large

TRHO Classification of bonds

	ρ	$Laplacian$	$\frac{\|V\|}{G}$	$\frac{H}{\rho}$	$\frac{\lambda_1}{\lambda_3}$
Covalent	$Large$	< 0	> 2	< 0	> 1
Transit (Non-covalent)	$Small$	> 0	$1 < \cdots < 2$	~ 0	
Ionic	$Small$	> 0	< 1	> 0	< 1

Other TRHO strategies

TRHO has many options to **search** the CPs (**global**, **specific**, **profile**). You can find the detailed list and their asociated parameters in the manual.

IAUTO	Strategy
-2	Search of CP given a **cluster/region** around an specific point
-1	Like **IAUTO=-2**, but considering **all the non-equivalent** atoms of the system
0	Search **between** unique atom pairs. Focused on BCPs.
1	Search given an **specific** starting set of **coordinates**
2	Search on the **straight line** connecting two atoms or two points.
3	Brute force approach using a **grid**.
4	Generates **profiles** for ρ, $\nabla^2\rho$, λ_3 and ϵ along the axis joining two atoms or two points.

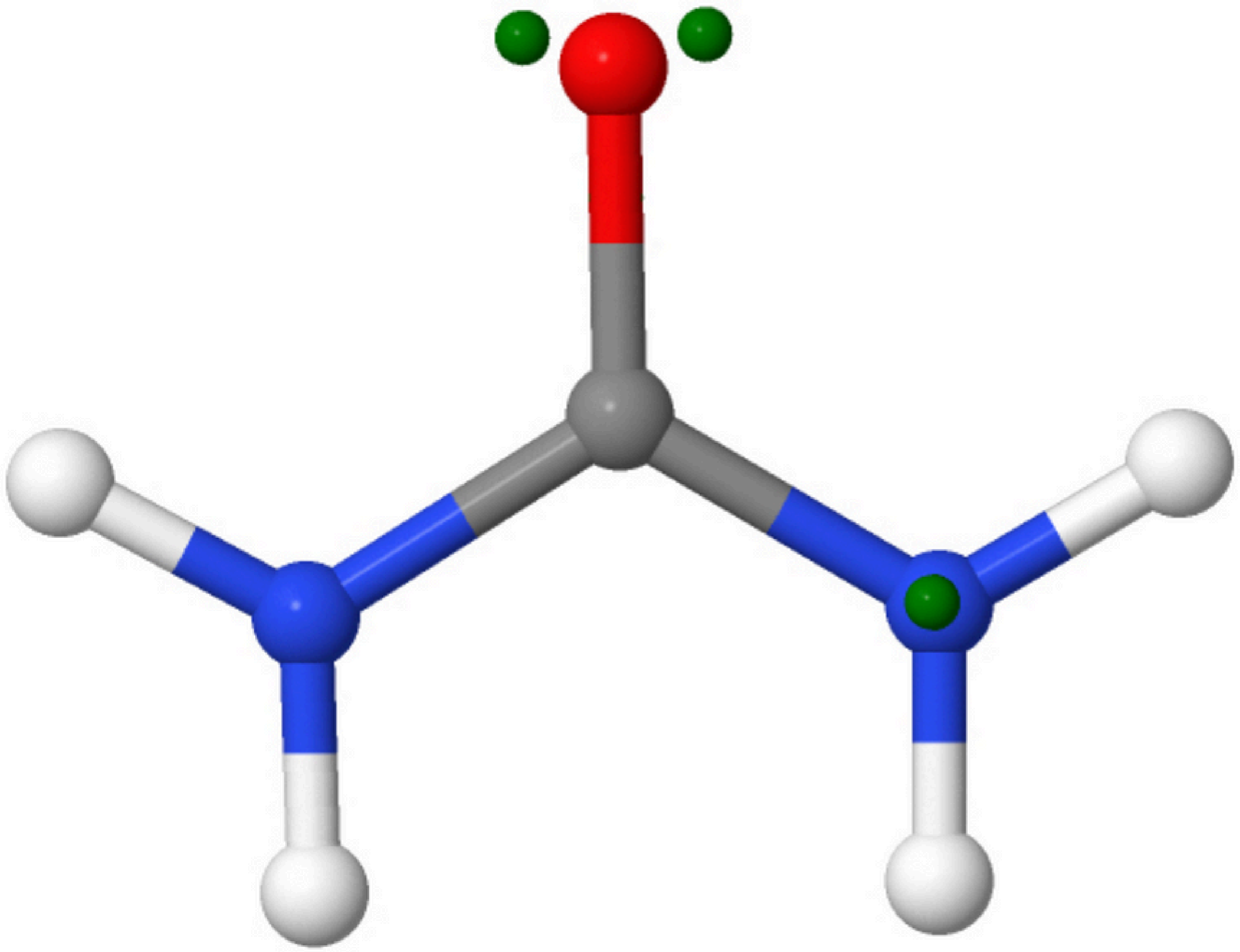
TRHO exercises

1. Build one table with the following quantities for one CP of each type:
   a. CP type
   b. Coordinates
   c. Properties and relevant quantities
   d. Eigenvalues and eigenvectors
   e. Ellipticity
   f. Length of the chemical bond
2. Build the same table for the Urea Molecule and compare.
3. Try to classify the nature of the chemical bond for the Urea Crystal
4. Try to classify the nature of the chemical bond for the Urea Molecule

EXTRA: Check the predicted interactions, given the obtained BCP, between a H2S molecule and the MOF NOTT-400 (follow the corresponding Notebook).

Topological Analysis of the Laplacian

- Keyword **TLAP**
- This second step concerns the topological analysis of the **Laplacian**, $\nabla^2\rho$
- The **atomic shell structure** is not visible through the topology of ρ, but it becomes generally so when that of $\nabla^2\rho$ is analysed, working like magnifying lens.
- When applied to atoms in chemical combination (like those in a crystal) the topology of $\nabla^2\rho$ reveals the **electron pairs** (shared, partially shared, or not shared lone pairs of the Lewis model)
- For the urea, we will search CPs in the **valence shell charge concentration** (VSCC) of the C, O, N and H atoms, respectively.



TLAP input

```
TOPO
TLAP
0
0,0,1,0,20,7,5
12,18
0
1
0,0
1
0,0
1
0,0
1
0,0.38
1
0,0.38
END
```

Keyword to call TOPOND

Keyword for TLAP functions

IAUTO: strategie to use

Parameters related to IAUTO=0. Explained in next slide

NT, NP: angular grid

NNA: number of NNA to be considered

For each non equivalent atom:

IYES: atom flag search

NMAX,RSTAR: standard search, default radius

TLAP input

- TOPOND on his core compute the **negative** of the Laplacian, $-\nabla^2\rho$
- Bonded or non-bonded **charge concentrations** (lone pairs) are in general associated to (3,-3) maxima in $-\nabla^2\rho$

IAUTO Strategies:

- IAUTO = 1: CPs search started from a given set of points
- IAUTO = 2: CPs search along the line joining nuclei A and B or two points a and b

- **IMETH**: Algorithm for searching (0 for NR, 1 for EF)
- **IEXT**: Choose to compute only properties depending on ρ and on its derivatives are evaluated or also on the non-diagonal elements of the first-order density matrix
- **IBPAT**: Atomic graph line lengths and termini are evaluated or not
- **IPRINT**: normal or debug printing
- **NSTEP**: max number of steps
- **NNB**: is the number of neighbors around each unique CPs
- **RMAX**: Maximum radius of sphere in the nearest neighbor analysis around each unique CP
- If IMETH = 1, insert the **ITYPE** to search only for one type of CP: 0 (3,-3); 1 (3,-1); 2 (3;+1); 3 (3,+3)

TLAP output

```
******************************************************************************

T O P O L O G I C A L   A N A L I S Y S   O F - (L A P L A C I A N   O F   R H O)

******************************************************************************


 COMPUTATIONAL CONDITIONS :
                          IAUTO                                    :     0
                          ALGORITHM FOR CP SEARCH                  :    NR
                          ATOMIC GRAPH EVALUATION                  :     T
                          MAX.NUM. OF N.R. OR EIG. FOL. STEPS      :    20
                          NEIGHBORS OF EACH NON-EQUIVALENT ATOM    :     7
                          PRINT N.RAPH OR E.FOL. SEARCH            :     F
                          NUMBER OF ANGLE THETA INTERVALS          :    12
                          NUMBER OF ANGLE PHI INTERVALS            :    18


 ****** CP SEARCH FOR NON-EQUIVALENT ATOM      1 C  ******

 SPHERE RADIUS (ANG)              :  4.9743E-01

 CP N.      1   VSCC OF NON-EQUIV. ATOM      1 C
 *********

 CP TYPE                          :  (3,-1)
 COORD(AU)  (X  Y  Z)             :  1.0478E+00  5.0523E+00  3.1918E+00
 COORD FRACT. CONV. CELL          :  9.9636E-02  4.8042E-01  3.6059E-01
 PROPERTIES (-LAP,GLAP,RHO)       : -1.1964E-01  1.9296E-15  1.3126E-01

 SPECTRAL DECOMP. OF THE HESSIAN OF -LAP(RHO)

 EIGENVALUES (L1 L2 L3)           : -3.5990E+00 -4.1033E-01  1.7021E+00
 EIGENVECTORS                     : -9.4526E-01 -2.2141E-02  3.2558E-01
                                     7.8014E-02  9.5343E-01  2.9134E-01
                                    -3.1687E-01  3.0079E-01 -8.9951E-01
```

Some important quantities to check:

- CP type
- CP Coordinates

```
 NON-EQUIV. ATOM C      1  NUMBER OF CPS FOUND: 24


     ********** C R I T I C A L   P O I N T S   F O U N D **********


  X(AU)   Y(AU)   Z(AU)    TYPE     -LAP     RHO    L1(V1)   L2(V2)   L3(V3)

 ******************************************************************************

  1.048   5.052   3.192  (3,-1)  -0.120    0.131   -3.599   -0.410    1.702
 ( C       1       0      0      0   1.111 AU )    -0.945   -0.022    0.326
                                                    0.078    0.953    0.291
                                                   -0.317    0.301   -0.900

  0.929   6.187   2.076  (3,-3)   1.583    0.436  -12.819  -10.916   -5.442
 ( N       5       0      1      0   0.999 AU )     0.625   -0.330    0.707
                                                    0.625   -0.330   -0.707
                                                   -0.467   -0.884    0.000

  0.000   5.258   4.397  (3,-3)   1.860    0.595  -40.770   -8.829   -7.528
 ( O       3       0      0      1   0.873 AU )     0.000   -0.707    0.707
                                                    0.000   -0.707   -0.707
                                                    1.000    0.000    0.000
```

TLAP Exercises

1. Build one table with the following quantities for one CP of each type :
   a. CP type
   b. Coordinates
2. Build the same table for the Urea Molecule and compare

Atomic Basins and their Properties

- Keyword **ATBP**
- A third step consists in the determination of **atomic basins** and in the consequent calculation of many one-electron properties by integration
- An atomic basin is defined as portion of space bounded by a surface whose points all fulfill the condition , where is a unit vector normal to the surface. The atomic surfaces are thus called **zero-flux surfaces**.
- In general every atomic property, like the electron population, the kinetic energy, the atomic multipole moments, etc. can be expressed in terms of a corresponding **three-dimensional density distribution** that can be integrated over ω to obtain its average value.

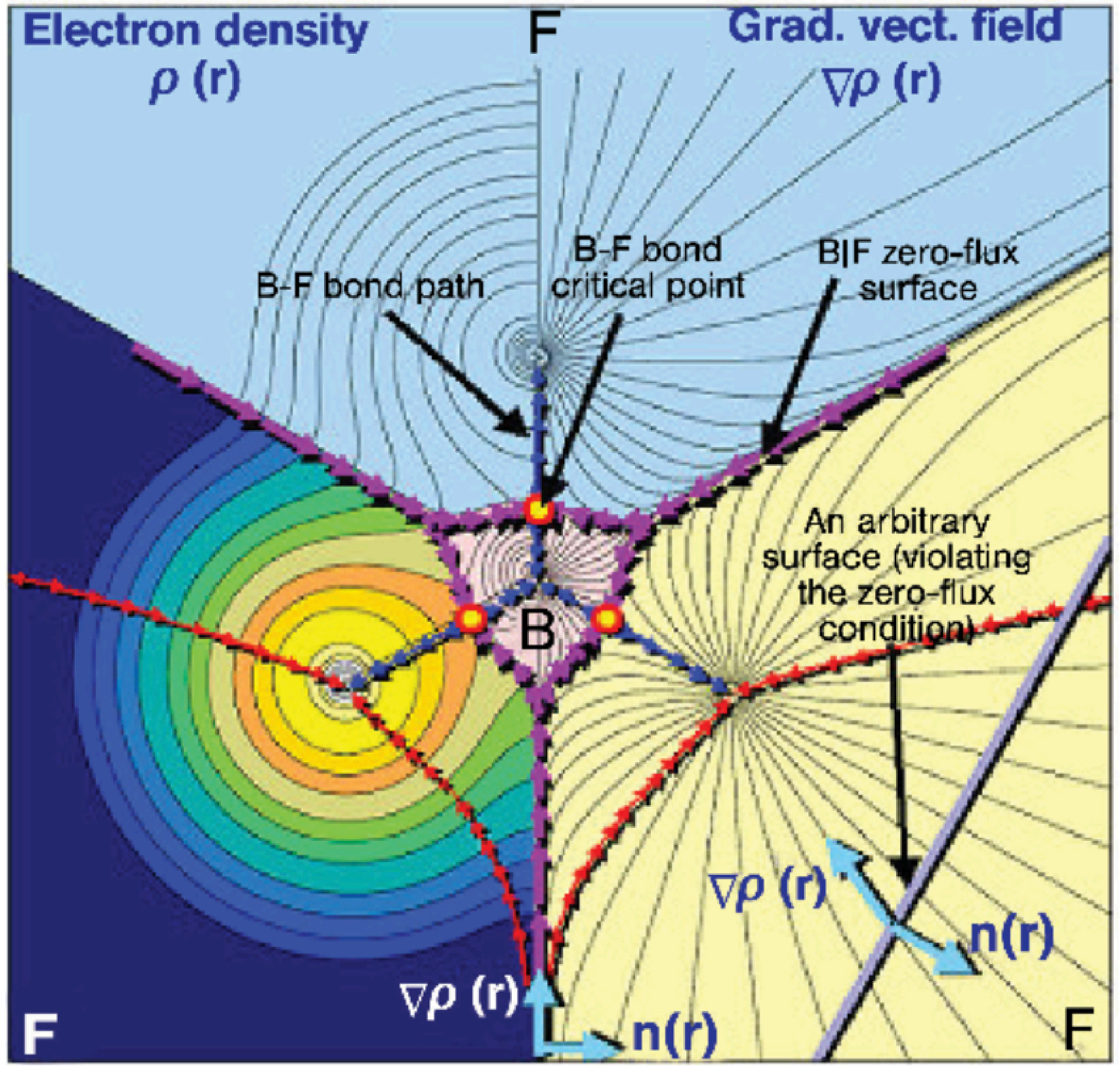
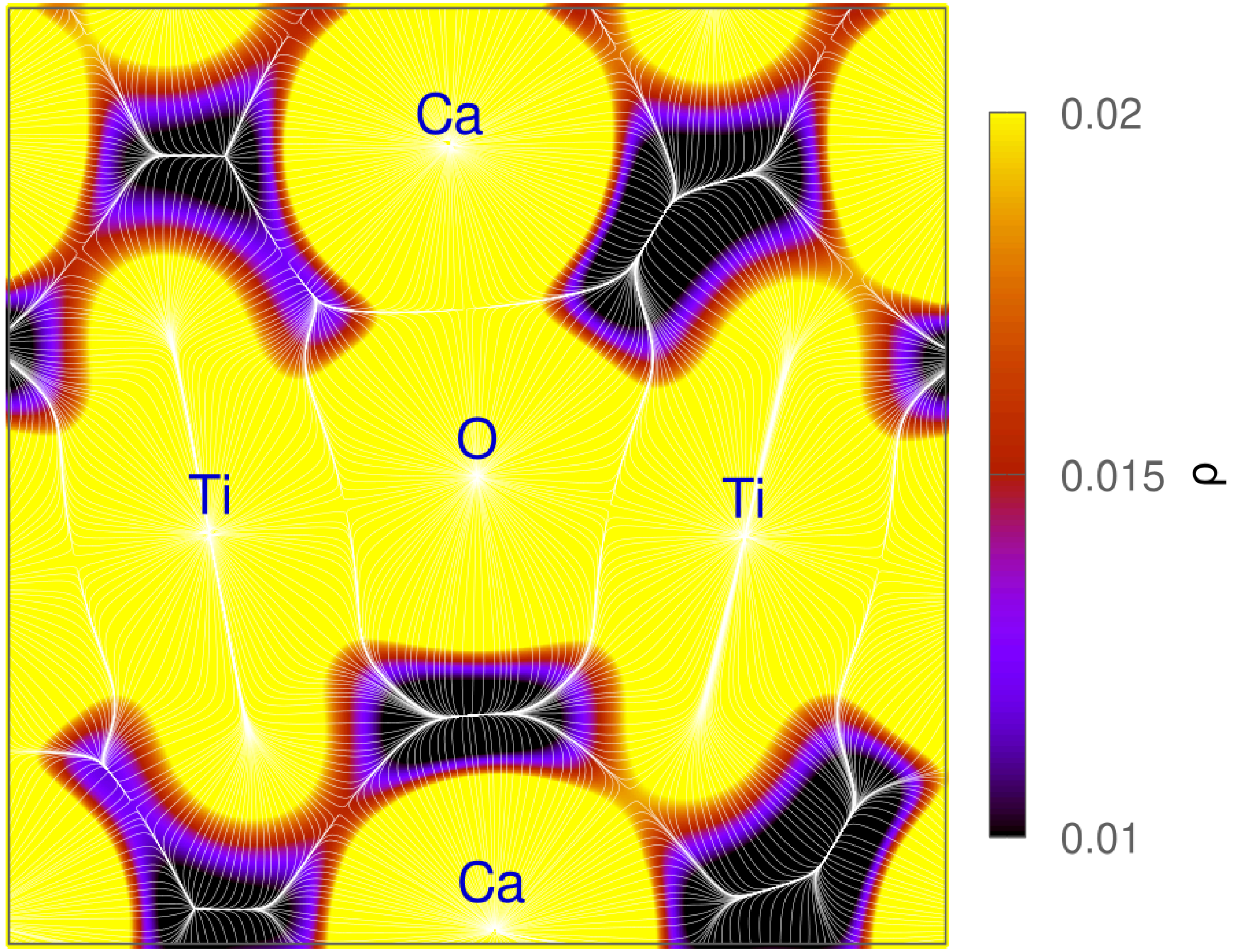


IMAGE from The Quantum Theory of Atoms in Molecules: From Solid State to DNA and Drug Design, C. Matta, R. J. Boyd, Wiley Online Library, 2007

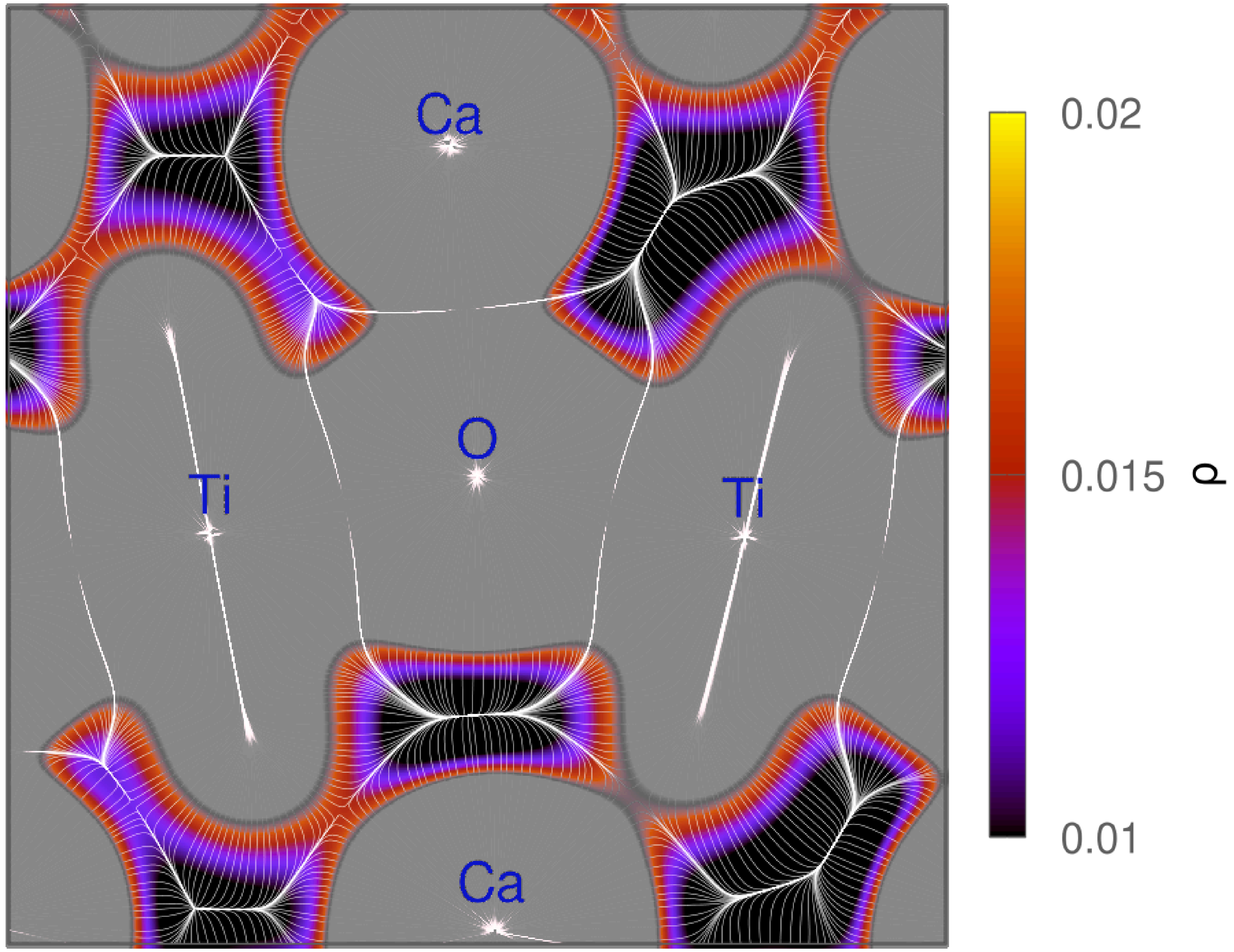
Atomic Basins and their Properties

- Keyword **ATBP**
- A third step consists in the determination of **atomic basins** and in the consequent calculation of many one-electron properties by integration
- An atomic basin is defined as portion of space bounded by a surface whose points all fulfill the condition , where is a unit vector normal to the surface. The atomic surfaces are thus called **zero-flux surfaces**.
- In general every atomic property, like the electron population, the kinetic energy, the atomic multipole moments, etc. can be expressed in terms of a corresponding **three-dimensional density distribution** that can be integrated over ω to obtain its average value.



Atomic Basins and their Properties

- Keyword **ATBP**
- A third step consists in the determination of **atomic basins** and in the consequent calculation of many one-electron properties by integration
- An atomic basin is defined as portion of space bounded by a surface whose points all fulfill the condition , where is a unit vector normal to the surface. The atomic surfaces are thus called **zero-flux surfaces**.
- In general every atomic property, like the electron population, the kinetic energy, the atomic multipole moments, etc. can be expressed in terms of a corresponding **three-dimensional density distribution** that can be integrated over ω to obtain its average value.



ATBP input

```
TOPO
ATBP
UNI
1, 0.43
10,0,1,0
32,24,120,2,1,0,0.002
1, 0.8
10,0,0,0
32,24,120,2,1,0,0.002
1, 0.74
10,0,0,0
32,24,120,2,1,0,0.002
1, 0.24
10,0,0,0
24,16,96,2,1,0,0.002
1, 0.24
10,0,0,0
24,16,96,2,1,0,0.002
0
END
```

Keyword to call TOPOND

Keyword for ATBP functions

UNI: integration performed for all the NEA

For each NEA:

IYES, TOL

NVI, IRSUR, IWSUR, IPRINT

IPHI, ITH, IBETP, IMUL, IEXT, NOSE, ACC

NNA

ATBP output

```
*************** A T O M I C   B A S I N   I N T E G R A T I O N ***************

COMPUTATIONAL CONDITIONS:
                          ATOM                                    : C     1
                          METHOD                                  :     0
                          NVI                                     :    10
                          IRSUR                                   :     0
                          IWSUR                                   :     1
                          NUMBER OF RADIAL POINTS INSIDE BETA SPHERE:   120
                          NUMBER OF PHI POINTS INSIDE BETA SPHERE  :    48
                          NUMBER OF THETA POINTS INSIDE BETA SPHERE :   32
                          NUMBER OF PHI POINTS OUTSIDE BETA SPHERE  :   32
                          NUMBER OF THETA POINTS OUTSIDE BETA SPHERE:   24
                          BETA SPHERE RADIUS (ANG)                :    0.42
                          SIZ (ANG)                               :    4.76
                          TIN (ANG)                               :    5.29

PROMEGA SURFACE STEP     :
                          STEP SIZE IN TRACING GRAD(RHO) TRAJEC.  :    0.025
                          ORDER OF A.B.M PREDICTOR CORRECTOR METHOD :   6
                          SMALL STEPS PER REGULAR STEP IN A.B.M.  :     5
                          PRECISION IN ATOMIC SURFACE DETERM. (AU) :   0.002
                          SECONDARY INTERSECTION SEARCH           :     N

ORIG.(AU)  (X  Y  Z)      :  0.0000E+00  5.2582E+00  2.8856E+00
```

```
ATOMIC POPULATIONS

           N    3.68683623637122E+00
           Q    2.31316376362878E+00

ATOMIC ENERGIES

           L    1.11399380533950E-02
           G    3.63030737459158E+01
           K    3.63142036326819E+01
          EG   -3.63567127948770E+01
          EK   -3.63678591264363E+01
        VNEO   -8.02808411061419E+01
       VNEOC   -8.03401499679325E+01

ATOMIC FORCES

        FAXA   -2.19698990303736E-03
        FAYA    1.02088976120362E-03
        FAZA   -3.27499037420670E-01

RADIAL ATOMIC EXPECTATION VALUES

       R(-1)    1.33801401843570E+01
       R(+1)    2.35695784880150E+00
       R(+2)    2.54549372459507E+00
       R(+3)    3.67056554672272E+00
       R(+4)    6.42181284353169E+00
      GR(-1)   -2.42506605804454E+01
      GR( 0)   -7.98456123670814E+00
      GR(+1)   -5.28894321642489E+00
      GR(+2)   -6.45458764841208E+00

ATOMIC VOLUMES AND RELATED POPULATIONS

        V001    2.19518073963665E+01
        N001    3.68578642040465E+00
        R001    9.99715252889127E-01
        V002    2.00924011012213E+01
        N002    3.68314075969884E+00
        R002    9.98997656409057E-01
        VTOT    2.32778786263969E+01

ATOMIC DIPOLE MOMENT VECTOR

          DX    6.20032581443473E-03
```

ATBP output

Populations		
Atomic population	N	$N(\Omega) = \int_\Omega \rho(\boldsymbol{r})d\tau$
Net charge	q	$q(\Omega) = Z(\Omega) - N(\Omega)$
α-Atomic population (α-electrons population)	N_A	$N_A(\Omega) = \int_\Omega \rho_\alpha(\boldsymbol{r})d\tau$
β-Atomic population (β-electrons population)	N_B	$N_B(\Omega) = \int_\Omega \rho_\beta(\boldsymbol{r})d\tau$
Excess of up-spin over down-spin atomic popultation	E_{SAP}	$E_{SAP} = N_A(\Omega) - N_B(\Omega)$
Energies		
Atomic Lagrangian: the error in L is a measure of the accuracy of the numerical integration	L	$L(\Omega) = -1/4 \int_\Omega \nabla^2 \rho\, d\tau$
Kinetic energy in terms of the dot product of the momentum operator → integration of the lagrangian kinetic energy density distribution	G	$G(\Omega) = 1/2 \int_\Omega (\boldsymbol{\nabla}\cdot\boldsymbol{\nabla}')\Gamma^{(1)}(\boldsymbol{r},\boldsymbol{r}')\|_{r=r'}\, d\tau =$ $= L(\Omega) = K(\Omega) - G(\Omega)$
Kinetic energy in terms of the Laplacian operator → integration of the hamiltonian kinetic energy density distribution	K	$K(\Omega) = -1/4 \int_\Omega (\nabla^2 + \nabla'^2)\Gamma^{(1)}(\boldsymbol{r},\boldsymbol{r}')\|_{r=r'} d\tau$
Atomic energy E=-E_{kin} corrected for the virial ratio	E_G E_K	$E_G(\Omega) = G(\Omega)\cdot(1+V/T)$ $E_K(\Omega) = K(\Omega)\cdot(1+V/T)$
Atomic value of nuclear-electron potential energy with its own nucleus; $\boldsymbol{R}_\Omega$ is the position vector of Ω in the system frame	V_{NEO}	$V_{NEO}(\Omega) = -\int_\Omega (Z_\Omega/r_\Omega)\,\rho(\boldsymbol{r})d\tau\ \boldsymbol{r}_\Omega =$ $= \boldsymbol{r} - \boldsymbol{R}_\Omega;\ r_\Omega \equiv \|\boldsymbol{r}_\Omega\|$
Atomic value of nuclear-electron potential energy with all nuclei in the system (only for molecular systems); $\boldsymbol{R}_B$ is the position vector of nucleus B in the molecular frame	V_{NET}	$V_{NET}(\Omega) = -\int_\Omega \sum_B (Z_B/r_B)\,\rho(\boldsymbol{r})d\tau\ \boldsymbol{r}_B =$ $= \boldsymbol{r} - \boldsymbol{R}_B;\ r_B \equiv \|\boldsymbol{r}_B\|$
Corrected values of nuclear-electron potential energies for the virial ratio	V_{NEOC} V_{NETC}	$V_{NEOC} = 2\cdot V_{NEO}\cdot(1+T/V)$ $V_{NETC} = 2\cdot V_{NET}\cdot(1+T/V)$

```
ATOMIC POPULATIONS

         N    3.68683623637122E+00
         Q    2.31316376362878E+00

ATOMIC ENERGIES

         L    1.11399380533950E-02
         G    3.63030737459158E+01
         K    3.63142036326819E+01
        EG   -3.63567127948770E+01
        EK   -3.63678591264363E+01
      VNEO   -8.02808411061419E+01
     VNEOC   -8.03401499679325E+01

ATOMIC FORCES

      FAXA   -2.19698990303736E-03
      FAYA    1.02088976120362E-03
      FAZA   -3.27499037420670E-01

RADIAL ATOMIC EXPECTATION VALUES

     R(-1)    1.33801401843570E+01
     R(+1)    2.35695784880150E+00
     R(+2)    2.54549372459507E+00
     R(+3)    3.67056554672272E+00
     R(+4)    6.42181284353169E+00
    GR(-1)   -2.42506605804454E+01
    GR( 0)   -7.98456123670814E+00
    GR(+1)   -5.28894321642489E+00
    GR(+2)   -6.45458764841208E+00

ATOMIC VOLUMES AND RELATED POPULATIONS

      V001    2.19518073963665E+01
      N001    3.68578642040465E+00
      R001    9.99715252889127E-01
      V002    2.00924011012213E+01
      N002    3.68314075969884E+00
      R002    9.98997656409057E-01
      VTOT    2.32778786263969E+01

ATOMIC DIPOLE MOMENT VECTOR

        DX    6.20032581443473E-03
```

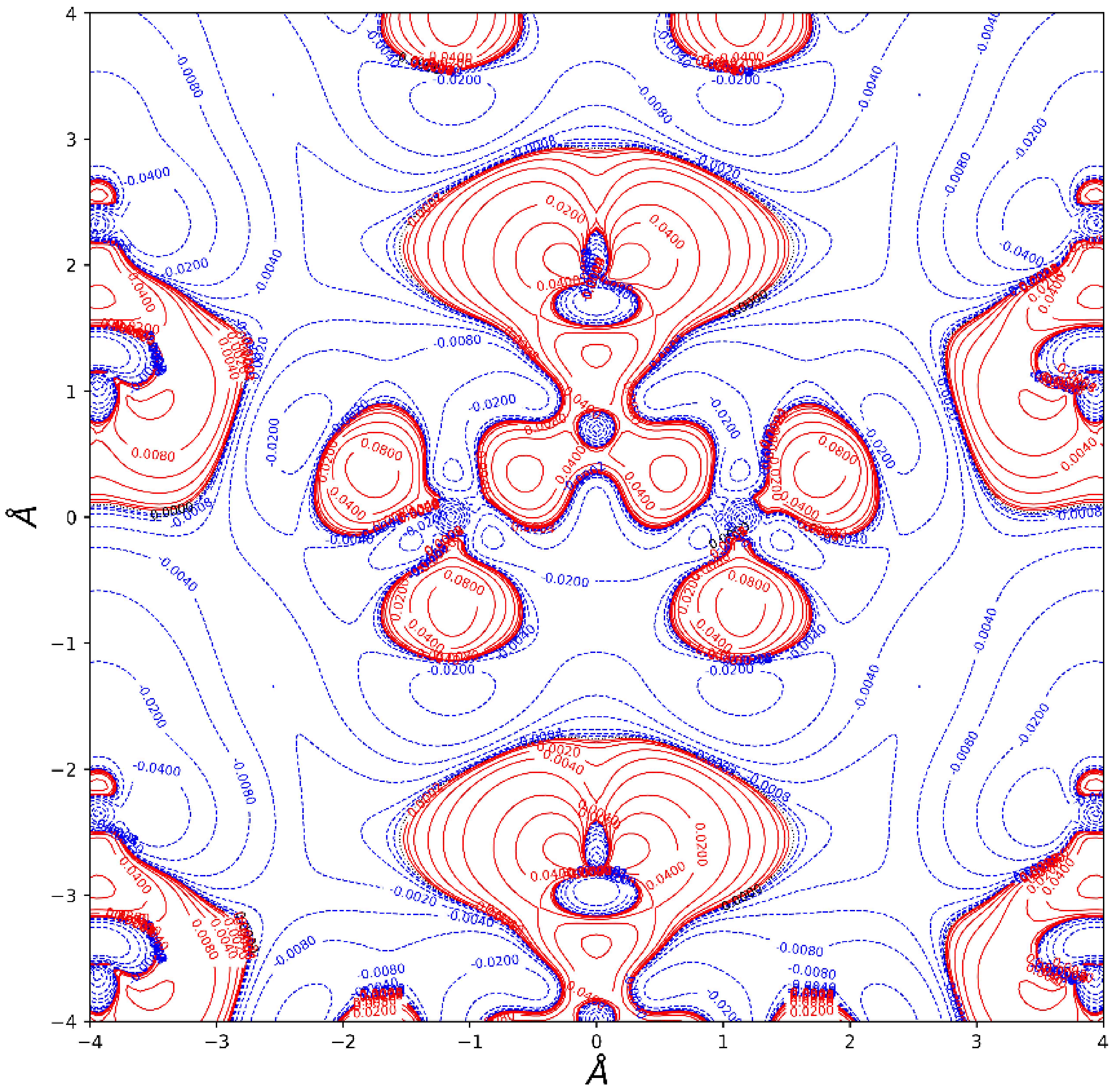
ATBP Exercises

1. Compare the following quantities for urea crystal and molecule:
    a. Atomic population
    b. Coordinates
    c. Atomic Lagrangian
    d. Kinetic energy
    e. Atomic Volume

Plotting Density (PL2D)

- Keyword **PL2D**
- A number of useful 2-dimensional **plots** on a plane defined through 3 points or nuclei or a suitable combination of points and nuclei is available in the TOPOND code.
- The 2D plots include **contour plots** of scalar functions and the tracing of gradient paths as partially summarized in the Table below:

Electron density	ρ
Spin density	$(\rho_\alpha - \rho_\beta)$
Laplacian of the electron density	$\nabla^2\rho$
Negative of the Laplacian of the electron density	$-\nabla^2\rho$
Magnitude of the gradient of ρ	$\lvert\nabla\rho\rvert$
Hamiltonian Kinetic energy density	K
Lagrangian Kinetic energy density	G
Virial field density	V
ELF (Becke electron localization function)	



PL2D input

```
TOPO
PL2D
6
0,-1,0
5
0,0,0
3
0,-1,1
3
0.5
30,15.0
1
-4,4,0.05
-4,4,0.05
1,1,1,1,1,1,1,1,1,1,1,1
Urea Crystal
0.55
2,0,0
2.2,1,1,1
36,0
END
```

Keyword for PL2D functions

For three atoms or coordinates:

KA: atom label of nucleus (0 if indicating coordinates)

LA, MA, NA: indices (direct lattice) of the cell where A is located.

If KA=0, indicate here the coordinates

ISHFT: choose where to put the origin

VMOD: shift the origin

NSTA, RMAX

ICONT: test run o normal run

XMI, XMA, XIN: Min, Max and grid interval along x-axis

YMI, YMA, YIN

IFU: if 1, a given property is evaluated

NAME: Plot title

SCALE: plot scale according to CRYSTAL output

TOLER, TR1, IPLANE: numeric tolerances

THR, IL (vector): maximum distance between bonded atomic pairs, trajectories that originate at BCP

NPATH, NEXTR: number of downhill trajectories originating from (3,-3) attractor, number of other attractors

PL2D output

```
 *** PLOT INFORMATION ***

 EULER ANGLES                        : 1.800E+02  9.000E+01  1.350E+02
 ROTAT. MATRIX                       : 7.071E-01  7.071E-01  0.000E+00
                                      -0.000E+00  0.000E+00  1.000E+00
                                       7.071E-01 -7.071E-01  0.000E+00
 COORD.,AT. N., CELL OF ATOMS A,B,C :-1.534E+00 -6.792E+00  1.563E+00      7      0    -1
0
                                       1.534E+00 -3.724E+00  1.563E+00      7      0     0

0
                                       0.000E+00 -5.258E+00  5.269E+00      8   0-1 1
 ORIGIN AT ( AU; SYSTEM REF. FRAME) : 0.000E+00 -5.258E+00  1.563E+00
 X AXIS RANGES AND INCREMENTS (ANG) :-4.000E+00  4.000E+00  5.000E-02
 Y AXIS RANGES AND INCREMENTS (ANG) :-4.000E+00  4.000E+00  5.000E-02
 IPSIZE(A3=1,A4=2), SCALE(ANG/CM)    :     2          5.500E-01
 TOLERA (ANG), TR1 (ANG), IPLANE     : 2.000E+00  1.000E-04    0
 THR (ANG), IL(K),K=1,3              : 2.200E+00   1   1   1
 N. GRHO TRAJ. FROM EACH 3,-3 CP     :36
 PLOTTED FUNCTIONS                   : ELECTRON DENSITY RHO
                                       SPIN DENSITY (RHOA-RHOB)
                                       LAPLACIAN OF RHO
                                       NEGATIVE OF THE LAPLACIAN OF RHO
                                       MAGNITUDE OF GRAD(RHO)
                                       HAMILTONIAN KINETIC ENERGY K(R)
                                       LAGRANGIAN KINETIC ENERGY G(R)
                                       VIRIAL FIELD DENSITY
                                       BECKE ELECTRON LOCALIZATION FUNCTION
                                       GRAD(RHO) TRAJECTORIES
                                       MOLECULAR GRAPH
                                       GRAD(RHO) TRAJECTORIES AND MOLECULAR GRAPH
```

File content	Name
$\rho(\boldsymbol{r})$	SURFRHOO.DAT
$\nabla^2\rho$	SURFLAPP.DAT
$-\nabla^2\rho$	SURFLAPM.DAT
$\lvert\boldsymbol{\nabla}\rho\rvert$	SURFGRHO.DAT
ELF	SURFELFB.DAT
V	SURFVIRI.DAT
G	SURFGKIN.DAT
K	SURFKKIN.DAT

PL2D Exercises

1. Try to plot one property using the plot_cry_contour function
2. Plot the same properties for the urea molecule and compare
3. Try to Plot a difference property

Thank you for your attention